

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_STAR CO_SC (MODEL-A & APEX) **Date: 10-01-2025**

Time: 3 Hrs **GTM-16(THEORY BASED-2)** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

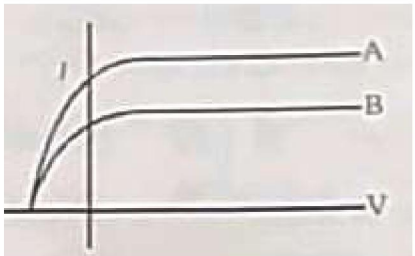
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A potential difference is applied across the ends of a metallic wire. If the potential difference is doubled, then the drift velocity:

   A) Will be doubled B) Will be halfed

   C) Will be quadrupled D) Will remain unchanged

2. The graphs show the variation of current I (y-axis) in two photocells, A and B as a function of the applied voltage V (x-axis) when light of same frequency is incident on the cell. Which of the following is the correct conculsion drawn from the data?



A) Cathodes of the two cells are made from the same substance and the intensity of light used are different

B) Cathodes are made from different substances and the intensity of light is the same

C) Cathode substances as well as intensity of light are different

D) No conclusion can be drawn

3. A linear isotropic dielectric slab is kept in an external electric field of strength E. If the magnitude of polarisation developed in the slab is P, then the electric susceptibility of dielectric slab is (ε_0 is the permittivity of free space)

A) $\frac{P}{\varepsilon_0 E}$ B) $\frac{P}{E}$ C) $\frac{E}{P}$ D) $\frac{\varepsilon_0 E}{P}$

4. A plane glass sheet is kept over various coloured pictures. The least raised picture is

A) Green B) Yellow C) Violet D) Red

5. In a reflecting telescope, a secondary mirror is used to

A) Remove spherical aberration

B) Reduce the problem of mechanical support

C) Make chromatic aberration zero

D) Move the eyepiece outside the telescope tube

6. One cannot see through fog, more appropriate answer is

A) Fog absorbs all the incident light falling on it

B) Light suffers total internal reflection at droplets

C) Refractive index of the fog is infinity

D) Light is scattered by the droplets

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7. **Assertion (A):** A small metal ball is charged to a potential of +1 V. It is introduced into a large hollow metal sphere charged to a potential of + 10,000 V and comes into contact with the inside surface of sphere. The ball's charge passes to the sphere completely.

Reason(R): Positive charge flows from a body at lower potential to another body at higher potential

A) both A and R are true and R is correct explanation of A

B) both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) both A and R are false

8. **Assertion (A):** In a microwave oven, porcelain container remains unaffected (do not get heated up) and cool.

Reason (R): Large number of molecules of porcelain vibrate and rotate with much smaller frequencies and cannot absorb microwaves.

A) both A and R are true and R is correct explanation of A

B) both A and R are true but R is not the correct explanation of A

C) A is true but R is false

D) both A and R are false

9. To manufacture a compound microscope of high magnification

A) Objective lens should be of high focal length

B) Objective lens should be of short focal length

C) Objective lens should be of larger aperture

D) Eyepiece should be of high focal length

10. Which of the following electromagnetic waves, vibrates not only electrons but also the entire atoms or molecules of a substance.

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A) Visible light B) UV light

C) Micro waves D) Infrared waves or heat waves

11. In the matching given below in column-I the type of electromagnetic waves are given and in column to their detection (method of detection / device / object used for detection) is given. Mark the option having correct matching

Column-I		Column-II	
1)	Micro wave	P)	Photo cells
2)	Infra – red	Q)	Ionisation chamber
3)	Light	R)	Point contact diodes
4)	x-rays	S)	Bolometer

A) 1-R, 2-S, 3-P, 4-Q B) 1-R, 2-Q, 3-P, 4-S

C) 1-P, 2-S, 3-R, 4-Q D) 1-S, 2-R, 3-P, 4-Q

12. For security reasons, a person at an airport is made to walk through the door way of a metal detector. If she/he is carrying anything made of metal, the metal detector emits a sound.On what principle does this detector work?

A) The metal detector works on the principle of resonance in AC circuits

B) The metal detector works on the principle of faradays law of electromagnetic induction

C) The metal detector works on the principle of Gauss's law in electrostatics

D) The metal detector works on the principle of Gauss's law in magnetism.

13. Induction furnace can be used to produce high temperatures and can be utilised to prepare alloys, by melting the constituent metals. Which of the following are responsible for producing high temperatures in metals

A) Fast moving electron beam incident on metals

B) X-ray beem illuminating on metals

C) Eddy currents generated in metals

D) Micro waves illuminating on metals

14. A: If a nucleus of mass number A =240 breaks into two A = 120 nuclei, nucleons get more tightly bound.

R: A very heavy nucleus say A =240 has high binding energy per nucleon compared to that of a nucleus with A =120.

A) both A and R are true and R is correct explanation of A

B) both A and R are true but R is not the correct explanation of A

C) A is true but R is false

D) both A and R are false

15. A: It is easier to remove a proton from $^{40}Ca_{20}$ nucleus as compared to a neutron.

R: Inside the nucleus, neutrons are acted upon only by attractive forces but protons are also acted upon by repulsive forces.

A) both A and R are true and R is correct explanation of A

B) both A and R are true but R is not correct explanation of A

C) A is true but R is false

D) both A and R are false

16. A: When charge inside a conducting cavity is increased then the force on charge kept outside the conductor increases.

R: Change in the charge inside conducting cavity changes the charge appearing on its outer surface.

A) both A and R are true and R is correct explanation of A

B) both A and R are true but R is not correct explanation of A

C) A is true but R is false

D) both A and R are false

17. Welding arc consists of large amount of

A) Visible light B) Infrared waves C) uv rays D) x-rays

18. A choke coil is in series with a lamp connected to a voltage source. When an iron core is inserted in the choke coil, compared to the initial steady state, in the final steady state, that brightness of lamp

1) increases if voltage source is DC source

2) Decreases if voltage source is AC source

3) Remains unchanged if voltage source is DC source

4) Remains unchanged if voltage source is AC source

A) 1 and 2 are correct B) 2 and 3 are correct

C) 1 and 4 are correct D) 3 and 4 are correct

19. For which of the following transparent media with respect to air, the critical angle is least

A) water B) crown glass C) Dense flint glass D) Diamond

20. The large scale transmission and distribution of electrical energy over long distances is done with use of transformers. The voltage output….X…..

It is then transmitted over long distances to an area sub-station near the consumers. There the voltage is…Y.. . It is further …Z…. at distributing sub-stations and utility poles before a power supply of 240 V reaches our homes.

In the above paragraph, fill the letters X,Y, & Z respectively with suitable operations

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A) Stepped –up. Stepped down and Stepped down

B) Stepped –up. Stepped down and Stepped up

C) Stepped –up. Stepped up and Stepped down

D) Stepped –down, Stepped up and Stepped down

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

NOTE: Question number's **21 to 25** rules for Answer marking. If your answer is options 2,4 Then you have to fill the OMR sheet as '24' If your answer is options 1,3,4 Then you have to fill the OMR sheet as '134' And if your answer is options 1,2,3,4 then you have to fill the OMR sheets as '1234'
If all the statements are incorrect then fill the OMR sheet as '0'

21. Which of the following statements are correct

1) Both the Thomson's as well as the Rutherford's models constitute an Unstable system.

2) Thomson's model is unstable electrostatically.

3) Rutherford's model is unstable because of electromagnetic radiation of orbiting electrons.

4) The orbital picture in Bohr's model of the hydrogen atom was inconsistent with the uncertainty principle.

22. Which of the following statements are correct about the motion of a particle

1) The shape of the trajectory of the projectile motion under gravity depends on the initial conditions of motion also.

2) A positive acceleration can be associated with a slowing down of a body.

3) An object moving along negative direction on a straight line with negative acceleration is slowing down.

4) Separation between two bodies moving in same direction on ground with same acceleration, will not increase.

23. Which of the following statements are correct regarding the polar satellites

1) The orbital radius of these satellites is nearly 6900 km to 7200 km.

2) They go around the poles of the earth in a north –south direction

3) The time period is around 100 minutes

4) These satellites can view, polar and equatorial regions at close distances with good resolution. Information gathered from such satellites is extremely used for remote sensing , meterology as well as for environmental studies of the earth.

24. In which of the following applications, eddy currents are used to advantage?

1) Electromagnetic damping in certain galvanometers

2) Magnetic breaking in trains

3) Heat produced in iron core of transformers

4) Electric power meters

25. Which of the following statements are correct related to moving coil galvanometer (the symbols have their usual meaning i e N = number of turns ; A = Area of each turn, B = external magnetic field & C = torsional constant)

1) The current sensitivity of galvanometer depends on N, A, B and C

2) The voltage sensitivity of galvanometer depends on N,A,B and C

3) When the current sensitivity is increased the voltage sensitivity also increases

4) When the current sensitivity is increased the voltage sensitivity may not necessarily increase.

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CHEMISTRY **MAX.MARKS: 100**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The compound which is used to remove plutonium as PuF_6 from spent nuclear fuel?

A) OF_2 B) O_2F_2 C) F_2 D) Cl_2O_7

27. Correct statement is

A) Hydronium ion has been detected in many compounds e.g: $H_3O^+Cl^-$ in solid state

B) In aqueous solution hydronium ion further hydrated to $H_3O_2^-, H_5O_3^-, H_7O_4^-$ etc

C) In aqueous solution hydroxyl ion further hydrated to $H_5O_2^+, H_7O_3^+, H_9O_4^+$ etc

D) In aqueous solution there is no evidence of existence of $H_3O_2^-, H_5O_2^+, H_5O_3^-$ & $H_7O_3^+$

28. Among Methylamine, Codeine, Quinine, Nicotine, Niacin, Phenolphthalein and Bromothymol blue, the no. of weak bases

A) 7 B) 6 C) 5 D) 4

29. Acid dissociation constant (Ka_2) values of Oxalic acid, Citric acid, Sulphurous acid, Phosphoric acid and Carbonic acid are A,B,C,D and E respectively, then correct order is

A) A > B > C > D > E B) B > A > C > D > E

C) B > C > A > D > E D) B > D > A > C > E

30. The following reaction is at equilibrium $Fe^{+3}_{(aq)} + SCN^-_{(aq)} \Leftrightarrow \left[Fe(SCN)\right]^{+2}_{(aq)}$ In the above reaction, the intensity of red colour can be increased by :

A) addition of oxalic acid(which reacts with Fe^{3+} ions)

B) addition of Hg^{2+} ions (which reacts with SCN^- ions)

C) addition of KSCN

D) removal of KSCN

31. Identify the incorrect $\left(i^* = Vant\ hoff's\ factor\ for\ incomplete\ dissociation\right)$

A) For $NaCl - i^*_{(0.1m)} > i^*_{(0.01m)} > i^*_{(0.001m)}$ B) For $K_2SO_4 - i^*_{(0.001m)} > i^*_{(0.01m)} > i^*_{(0.1m)}$

C) For $MgSO_4 - i^*_{(0.1m)} < i^*_{(0.01m)} < i^*_{(0.001m)}$ D) For $KCl - i^*_{(0.1m)} < i^*_{(0.01m)} < i^*_{(0.001m)}$

32. Mixture of Phenol & Aniline shows negative deviation because

A) Intermolecular Hydrogen bonding in Phenol

B) Intermolecular Hydrogen bonding in Aniline

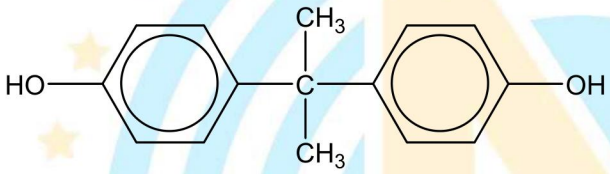
C) The inter molecular hydrogen bonding between phenolic proton and lone pair of nitrogen atom of aniline is stronger

D) Stronger Hydrogen bonding in Aniline

33. Which of the following is Incorrect?

A) Λ_m – increases with dilution; E^{Θ}_{Cell} – intensive property; Conductivity - depends on number of ions per volume; $\Delta_r G_{Cell}$ – extensive property.

B) Corrosion can be prevented by covering the surface with



(Bisphenol)

C) Polyacetylene when exposed to vapours of iodine acquires metallic lustre and conductivity.

D) Conductivity of 0.1M KCl greater than 0.1M HCl

34. Correct match of the following

	Column-I	Column-II (Space between electrodes is filled with)
A)	Leclanche cell	P) Moist paste of $NH_4Cl + ZnCl_2$
B)	Mercury cell(Cell potential remains constant during its life)	Q) paste of KOH+ZnO
C)	Lead storage cell	R) 38% H_2SO_4
D)	Ni-Cd cell(longer life than lead storage cell)	S) Moist NaOH (or) KOH

A) A-P, B-Q, C-S, D-R B) A-P, B-Q, C-R, D-S

C) A-R, B-P, C-R, D-S D) A-P, B-R, C-Q, D-S

35. The traces present in Misch metal along with lanthanoid metal & Iron

A) S only B) C only C) Ca only D) S,C, Ca & Al

36. Which of the following statement is FALSE.

A) During SN^1 reaction, the nucleophile plays no kinetic role in the rate of reaction.

B) During SN^2 reaction, if the substrate is optically pure with (R)- configuration, then the product obtained is always an optically pure product with (S) – configuration.

C) SN^2 reaction takes place in single step and no intermediate is formed.

D) The boiling points of isomeric alkyl halides decreases with increase in branching.

37. Which of the following statement is FALSE

A) The boiling point of 2-Butanol is less than 1-Butanol

B) The boiling point of methoxymethane is more than propane

C) The pKa order of various degree isomeric alcohols follows, primary alcohols > secondary alcohols > tertiary alcohols

D) Commercially carboxylic acids are reduced to alcohols by converting them to esters first, followed by reduction using hydrogen rather than reducing the carboxylic acids directly to alcohols using lithium aluminium hydride, because $LiAlH_4$ is an expensive reagent

38. Which of the following statement is FALSE

A) Salicylaldehyde and Cinnamaldehyde are compounds having pleasant fragrances

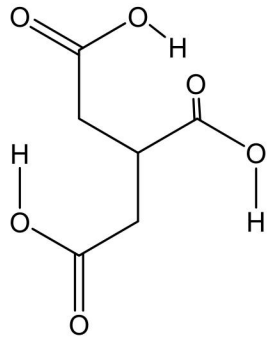
B) Vanillin structure contains an aldehyde, Ether and alcohol functional groups.

C) During the oxidation of α,β – unsaturated ketones to corresponding carboxylates using sodium hypohalite the carbon – carbon double bond does not get effected.

D) Formalin, a 40% solution of formaldehyde is used to preserve biological specimens

39. Which of the following is FALSE

A) The common name of the following compound is carballylic acid



B) Aliphatic carboxylic acids upto nine carbon atoms are colourless liquids at room temperature.

C) Simple aliphatic carboxylic acids upto four carbon atoms are miscible in water.

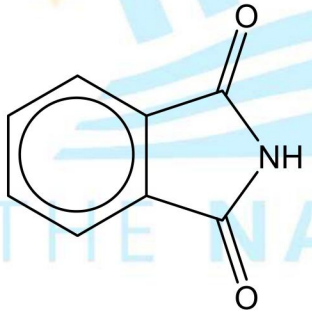
D) Benzoic acid is completely soluble in cold water.

40. Which of the following statement is FALSE

A) Cyclohexanone forms cyanohydrin in good yield but 2,2,6-trimethyl cyclohexanone does not

B) During the preparation of esters from carboxylic acid and an alcohol in the presence of acid catalyst, the water or ester should be removed as soon as it is formed.

C) The common name of the following compound is phthalamide



D) The acidic strength of $\overset{Br}{\underset{|}{CH_2}}-COOH$ is greater than $\overset{Cl}{\underset{|}{CH_2}}CH_2-COOH$

41. Which of the following statement is true

A) Adrenaline is a nitrogen compound contains tertiary amine functional group

B) Benadryl a well known antihistamine drug contains secondary amine functional group

C) Ephedrine contains secondary amine functional group

D) Butan-1-amine is more soluble in water than that of Butan-1-ol

42. Which of the following statement is FALSE

A) Direct nitration of aniline yields tarry oxidation products in addition to nitro derivatives

B) Nitration of aniline with nitration mixture at 288K yields ortho nitro aniline as minor product among the three isomeric nitro anilines

C) Sulphanilic acid exists as zwitter ion

D) Benzene diazonium fluoroborate is water soluble

43. Which of the following statement is False

A) In benzene diazonium ion, the positive charge can be delocalised in benzene ring during resonance

B) Alkyl diazonium salts are highly unstable, whereas Arene diazonium salts are stable for short time in solution at low temperatures

C) Benzene diazonium chloride is colourless solid and is readily soluble in water

D) Benzene diazonium fluoroborate when heated in the presence of aqueous sodium nitrite and copper gives fluorobenzene and this reaction is called Balz-Schiemann reaction.

44. Which of the following statement is false

A) Glucose fails to give Schiff's test

B) Glucose reacts with $NaHSO_3$ and gives white crystalline addition product

C) Glucose reacts with hydroxyl amine but penta acetate of glucose does not reacts with hydroxyl amine.

D) The boiling point and melting points of $\alpha-D$ glucose is less than $\beta-D$ glucose

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45. Which of the following statement is False

A) Glycine is sweet in taste

B) α – Helix structure of protein in which polypeptide chain – forms all possible hydrogen bonds by twisting into a left handed screw with – NH – groups of each amino acid residue hydrogen bonded to an adjacent turn of the helix.

C) Curdling of milk is an example of denaturation

D) Tyrosine was first – obtained from cheese.

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46. In dilute solutions molarity is less than X M, the activity of hydrogen ion $\left(a_{H^+}\right)$ is equal to magnitude of molarity of hydrogen ion $\left[H^+\right]$. If the solubility of the salt less than Y M, then it is sparingly soluble, then $\frac{X}{Y}$ value is(where M is molarity)

47. The no.of correct statements is ____

(1) VX_5 undergo hydrolysis gives VOX_3(X = Cl, Br)

(2) The +7 state for Mn is not observed in simple halides but it is possible in MnO_3F

(3) The tetrahedral $[MO_4]^{n-}$ ions are known for V^V, Cr^{VI}, Mn^V, Mn^{VI} and Mn^{VII}

(4) If a non-metal is like H, C or N are trapped inside the crystal lattices of metals, they becomes usually non-stoichiometric

(5) If a non-metal is like H, C or N are trapped inside the crystal lattices of metals, they becomes neither typically ionic nor covalent

(6) If a non-metal is like H, C or N are trapped inside the crystal lattices of metals, they exhibits fractional oxidation states

48. The no.of correct statements is _____

(1) The structural variability of actinoids is obtained due to irregularities in metallic radii which are far greater than in lanthanoids.

(2) The +3 and +4 ions tend to hydrolyse in actinoids because the distribution of oxidation states is so uneven and so different than lanthanoids

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(3) The magnetic properties of actinoids are more complex than those of the lanthanoids, though variation in the magnetic susceptibility of the actinoids with the number of unpaired 5f electrons is roughly parallel to the corresponding results for lanthanoids

(4) Only Pu shows maximum oxidation state of +7 in actinoids

(5) M^{4+} ion of Th is the only diamagnetic M^{4+} ion of actinoid series

(6) Electrons present in the 5f orbitals of actinides can more effectively shielded from nuclear charge as compared to electrons present in 4f orbitals of lanthanides

49. The no.of Correct orders/statements is _____

(1) $N < P < As < Sb < Bi$ (MP order)

(2) $BiH_3 > SbH_3 > NH_3 > ASH_3 > PH_3$ (BP order)

(3) $S > Se > Te > O$ $\left(\left|\Delta_{eg}H\right|\left(Electron\ gain\ enthalpty\right)\right)$

(4) HI > HF > HBr > HCl (MP order)

(5) Among $S_2Cl_2, Se_2Cl_2, S_2Br_2, Se_2Br_2, H_3PO_4, H_3PO_3, H_3PO_2$ which undergo disproportionation on heating $S_2Cl_2, Se_2Cl_2, S_2Br_2$ only

(6) Among Hydrogen halides, only HF exist as liquid at room temperature

50. The no.of Correct statements

(1) The compound of radon identified by radiotracer technique is RnF_2

(2)) Gold combines with cyanide to the presence of oxygen and water to form the coordination entity $\left[Au(CN)_4\right]^-$

(3) In the refining of nickel, impure nickel is converted to $\left[Ni(CO)_5\right]$, which is decomposed to yield pure nickel

(4) In black and white photography, the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion $\left(Ag(S_2O_3)_2\right)^{-3}$

(5) Articles can be electroplated with silver and gold form solution of the complexes

$\left[Ag(CN)_3\right]^-$ and $\left[Au(CN)_2\right]^-$

(6) The excess of copper and iron in plant/animal system are removed by chelating ligands D – pencillamine and disferrioxime – B respectively.

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. Assertion (A): The lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $\frac{9x-16}{13}=\frac{9y-1}{7}=\frac{z}{-1}$ are coplanar

Reason (R): Two lines $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$ are coplanar if $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ a_2-a_1 & b_2-b_1 & c_2-c_1 \end{vmatrix}=0$.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

52. Assertion (A): If the direction cosines of three perpendicular lines are $(\ell_1,m_1,n_1),(\ell_2,m_2,n_2),(\ell_3,m_3,n_3)$ then direction cosines of a line which is equally inclined with them are $\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}},\frac{m_1+m_2+m_3}{\sqrt{3}},\frac{n_1+n_2+n_3}{\sqrt{3}}$.

Reason (R): The cosines of the angle between the last line and any of the given lines is $\frac{1}{\sqrt{3}}$.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

53. Assertion (A): The shortest distance between skew lines $\frac{x+3}{-4}=\frac{y-6}{3}=\frac{z}{2}$ and $\frac{x+2}{-4}=\frac{y}{1}=\frac{z-7}{1}$ is 9.

Reason (R): Two lines are skew lines if there exists no plane containing them

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

54. Assertion (A): For $2 < a < 5$, $\int\frac{dx}{x^2+2(a-3)x+(a-1)}=A\ln\left|g(x)\right|+C$; where A is constant and C is integration constant, $g(x)=\left|\frac{(x+a-3)-f(a)}{(x+a-3)+f(a)}\right|$ and $f(a)=\sqrt{a^2-7a+10}$

Reason (R): For $2 < a < 5$, $\frac{1}{x^2+2(a-3)x+(a-1)}$ is a continuous function.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

55. Assertion (A): $\int_{-\pi}^{\pi}\frac{2x\left(1+\sin^2 x\right)}{1+\cos^2 x}dx=0$

Reason (R): $\int_{-\pi/2}^{\pi/2}\sin^2 x\cos^2 x(\sin x+\cos x)dx=\frac{4}{15}$

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

56. The solution of the differential equation

$$\frac{y+\sin x.\cos^2(xy)}{\cos^2(xy)}dx+\left(\frac{x}{\cos^2(xy)}+\sin y\right)dy=0 \text{ is}$$

A) $\tan xy+\cos x+\cos y=c$ B) $\tan xy-\cos x-\cos y=c$

C) $\tan xy+\cos x-\cos y=c$ D) $\tan xy-\cos x+\cos y=c$

57. Assertion (A): The area bounded by the curves $y=x^2+2x-3$ and the line $y=\lambda x+1$ is least, if $\lambda=2$.

Reason (R): The area bounded by the curve $y=x^2+2x-3$ and $y=\lambda x+1$ is $\left\{(\lambda-2)^2+16^{3/2}\right\}$ square units.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

58. S_1: The matrix B^TAB is symmetric or skew-symmetric according as A is symmetric or skew-symmetric matrix.

S_2: If A and B are symmetric matrices of the same order then $AB-BA$ is a symmetric matrix.

S_3: Null matrix of order $n\times n$ satisfies both the symmetric and skew-symmetric property

Then the number of statements which are true

A) 1 B) 2 C) 0 D) 3

59. Statement – I: $f:R\to R$ be a function such that $|f(x)|\le x^2,\forall x\in R$ then $f(x)$ is differentiable at $x=0$.

Statement – II: $f:R\to R$ be a function such that $|f(x)|\le x^p,\forall x\in R$ then $f(x)$ is differentiable at x = 0 for any whole number 'p'

A) Both statement – I and statement II are true

B) Both statement- I and statement II are false

C) Statement I is true and statement II is false

D) Statement I is false and statement II is true

60. Let f(x) be continuous on [a, b] and differentiable on (a, b)

Number of correct statements among the following.

Statement I: If f(x) is strictly increasing on (a,b) then $f^1(x) \geq 0$ for all $x \in (a,b)$

Statement-II:If $f(x)$ is strictly decreasing on (a,b) then $f^1(x) < 0$ for all $x \in (a,b)$

Statement-III: $f(x)$ and $f^1(x)$ have opposite sign, for all x, then $f^2(x)$ is a decreasing function

Statement-IV: $f(x)$ and $f^1(x)$ have opposite sign, for all x, then $|f(x)|$ is an increasing function

A) 1 B) 2 C) 3 D) 4

61. Let the function $f : R \to R$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g : R \to R$ be an arbitrary function.

Let $fg : R \to R$ be the product function defined by $(fg)(x) = f(x)g(x)$

Then number of correct statements among the following

1) If g is discontinuous at x = 1, then fg can never be differentiable at x = 1

2) If fg is differentiable at x = 1, then g is continuous at x = 1

3) If fg is differentiable at x = 1, then g must be differentiable at x = 1

A) 3 B) 2 C) 1 D) 0

62. Let $A = \{0,1,2,3,4,5,6,7\}$. Then the number of bijective functions $f : A \to A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to ____

A) 700 B) 620 C) 710 D) 720

63. Which of the following must be the true value of the statements below in that order.

Statement I: For square matrices A and B of same order. (with real entries) (O is null matrix and I is Identity matrix of same order as A and B). If AB = O then $\left(\det A\right)^2+\left(\det B\right)^2=0$

Statement II: If l_i,m_i,n_i (i = 1, 2, 3) denote the direction cosines of 3 mutual perpendicular linessuch that a matrix $A=\begin{pmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{pmatrix}$ then det (A) = ±1

(det (A) represents determinant of a matrix A)

Statement III: For two distinct Lines with shortest distance $d \neq 0$ we can have a point A online one and point B on line two such that AB=d then A and B be unique points on them inall possible cases

A) FFT B) FFF C) FTF D) FTT

64. Consider the system of linear equations $-x+y+2z=0, 3x-ay+5z=1, 2x-2y-az=7$

Let S_1 be the set of all $a \in R$ for which the system is inconsistence and S_2 be the set of all $a \in R$ for which the system has infinitely many solutions. If $n\left(S_1\right)$ and $n\left(S_2\right)$ denote the number of elements in S_1 and S_2 respectively then

A) $n\left(S_1\right)=2, n\left(S_2\right)=2$ B) $n\left(S_1\right)=1, n\left(S_2\right)=0$

C) $n\left(S_1\right)=2, n\left(S_2\right)=0$ D) $n\left(S_1\right)=0, n\left(S_2\right)=2$

65. Let $\vec{a},\vec{b}$ be two non collinear unit vectors and $\vec{x}=\vec{a}-\left(\vec{a}.\vec{b}\right)\vec{b}$ and $\vec{y}=\vec{a}\times\vec{b}$. Now consider the following statements:

Statement-I: $\left|\vec{x}\right|=\left|\vec{y}\right|$

Statement-II: $\left|\vec{y}\right|=\left|\vec{x}\right|+\left|\vec{x}.\vec{b}\right|$

A) Statement-I is true but Statement-II is false

B) Both Statement-I and Statement-II are false

C) Both Statement-I and Statement-II are true

D) Statement-I is false but Statement-II is true

66. The distance between the two parallel lines is 1 unit. A point 'A' is chosen to lie between the lines at a distance 'd' from one of them. Triangle ABC is equilateral with B on one line and C on the other parallel line. The length of the side of the equilateral triangle is

A) $\frac{2}{3}\sqrt{d^2+d+1}$ B) $2\sqrt{\frac{d^2-d+1}{3}}$ C) $2\sqrt{d^2-d+1}$ D) $\sqrt{d^2-d+1}$

67. Lines $L_1: y-x=0$ and $L_2: 2x+y=0$ intersect the line $L_3: y+2=0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement-1: The ratio PR : RQ equals $2\sqrt{2}:\sqrt{5}$

Statement-2: In any triangle, bisector of an angle divides the triangle into two similar triangles

A) Statement-1 is true, statement-2 is true; statement-2 is a correct explanation for statement-1

B) Statement-1 is true, statement-2 is true; statement-2 is NOT a correct explanation for statement-1

C) Statement-1 is true, statement-2 is false

D) Statement-1 is false, statement-2 is true

68. Let the solution curve y = f(x) of the differential equation $\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{15x^4+4x^3+4x+1}{\sqrt{1-x^2}}, x\in(-1,1)$ passes through the origin. Then $\int_{-1/2}^{1/2}\frac{f(x)dx}{(x^2+2)}$ is equal to

A) $\frac{\pi}{6}$ B) $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$ C) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ D) $\frac{\pi}{3}-\frac{1}{4}$

69. Let $S_k=\sum_{r=1}^{k}\tan^{-1}\left(\frac{6^r}{2^{2r+1}+3^{2r+1}}\right)$. Then $\lim_{k\to\infty}S_k$ is equal to

A) $\tan^{-1}\left(\frac{3}{2}\right)$ B) $\frac{\pi}{2}$ C) $\cot^{-1}\left(\frac{3}{2}\right)$ D) $\tan^{-1}(3)$

70. The value of the definite integral $\int_{\pi/8}^{3\pi/8}\frac{11+4\cos 2x+\cos 4x}{1-\cos 4x}dx$ equals

A) $-6-\frac{\pi}{4}$ B) $6\sqrt{2}-\frac{\pi}{4}$ C) $12-\frac{\pi}{2}$ D) $6-\frac{\pi}{4}$

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SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. The function $f: N \times N \to R$ is defined by

$$f(i,i+1) = \frac{1}{3} \forall i$$

$$f(i,j) = f(i,k) + f(k,j) - 2f(i,k)f(k,j) \forall k \in (i,j) \text{ such that } f(1,100) = \frac{1}{2}\left(1 - \frac{1}{3^k}\right)$$

then the value of k =

72. The number of solutions of the equation $\sin^2 x + (2 + 2x - x^2)\sin x - 3(x-1)^2 = 0$ in $\left[0, \frac{\pi}{2}\right]$ is α, and in $[-2\pi, 2\pi]$ is β then $\alpha + \beta =$ ____

73. $\lim\limits_{n\to\infty} \dfrac{\sum\limits_{k=1}^{n-1}(k-1)(nk-k^2)}{2\sum\limits_{r=1}^{n} r^3 - \sum\limits_{s=0}^{n}\left(s^2 + (n-s)^2\right)} = t$ then $[43t] =$ _____ ([.] denotes GIF)

74. $xy^2 = 100$ with $1 \le x \le 10$. The smallest value of $(\log_{10} x)^2 + (\log_{10} y)^2$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers,then $m + n$.

75. If $\left[\sqrt{1}\right] + \left[\sqrt{2}\right] + \left[\sqrt{3}\right] + ... + \left[\sqrt{2016}\right]$ = k, then $\left[\frac{k}{59}\right] =$ _____.

[.] denotes greatest integer function



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